

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000810	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	28-Jun-2011
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DG-57 / DG-517 Service Manual

SAMPLES CAMERA ADJUSTMENT

Procedure description
This procedure describes how to adjust Samples camera.

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EQUIPMENT

1. An instrument connected with a PC with the software installed.
2. Tubes of 15mm, 12mm and 10mm diameter with user's labels and barcodes.

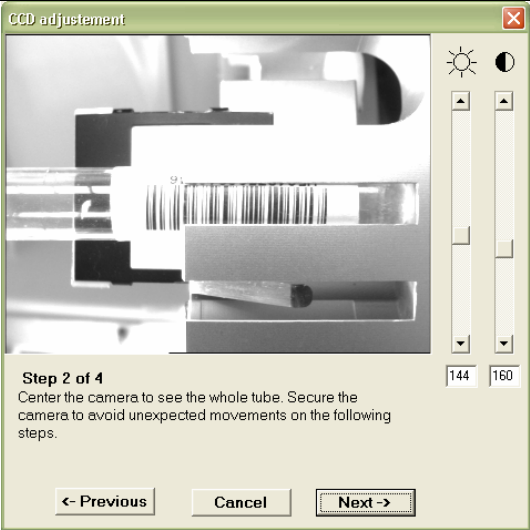
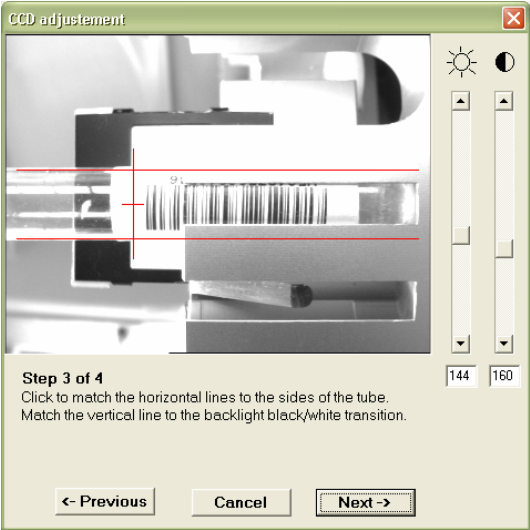
SET UP

1. Turn on the instrument.
2. Start the ST software (Technical Service software).
3. Enter the right Technical Service password (d+g_diana).

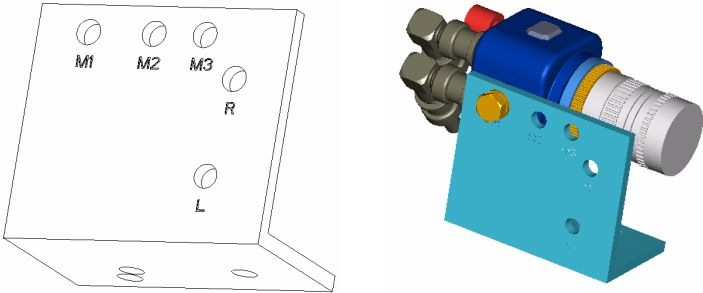
PROCEDURE

Process	Expected Results
Adjusting Samples camera position 1. In <i>Settings</i> tab, click on <i>Samples</i> button in <i>Camera alignment</i> section  Picture 1 2. Follow the instructions on the screen. Place a 15mm tube in position 1 of Samples Carousel.  Picture 2 3. Click on <i>Next</i>. 4. Move the camera up/down and right/left to center the whole tube in the image. Lock the camera position.	New window appears (Picture 2). --- New window appears (Picture 3). ---

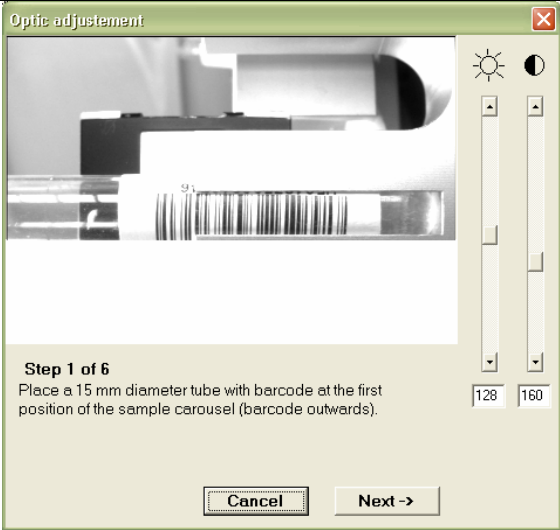
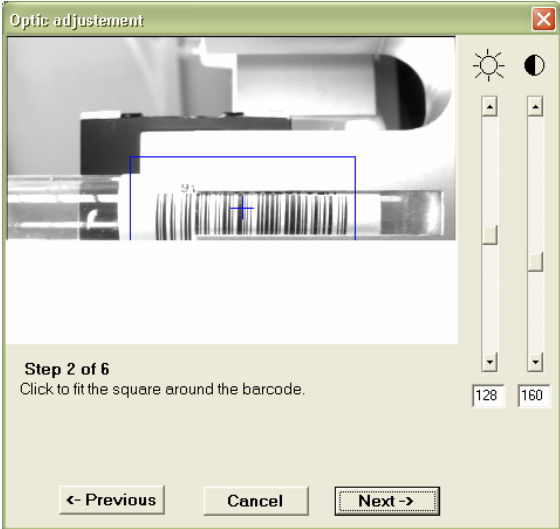
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Process	Expected Results
 Picture 3 - Click on <i>Next</i>. - Center the lines to the position shown in the image. The vertical line must fit the top of the electroluminescent screen. The two horizontal lines have to be centered to the tube.  Picture 4 - Click on <i>Next</i>. A new window appears. Click on <i>Finish</i>. - The Samples camera holder has 3 possible positions for the camera (see Picture 5).	New window appears (Picture 4). --- ---

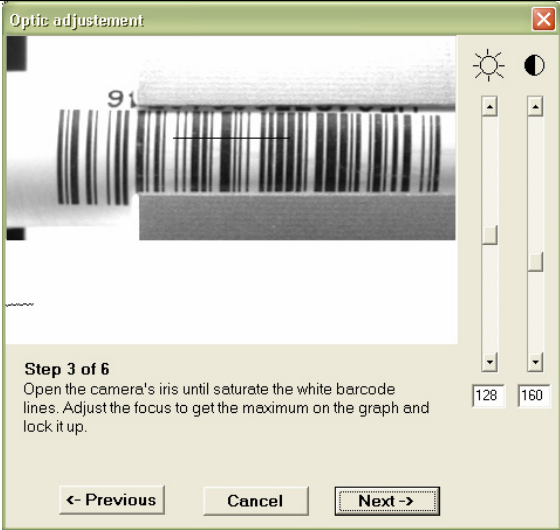
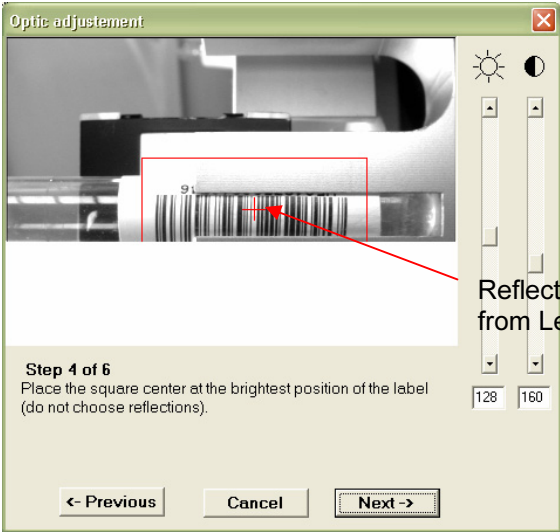
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Process	Expected Results
 Picture 5 Type manually the distance (in mm.) in the textbox <i>Camera distance</i> according to the position of the camera (see Picture 6):  Picture 6 Camera distance for the three positions must be: ▪ M1=148 mm ▪ M2=130 mm ▪ M3=116 mm Adjusting Samples camera optics - Disable temperature control: access <i>I/O</i> tab and click on <i>Off</i> checkbox in <i>Heating control</i> section. - In <i>Settings</i> tab, click on <i>Samples</i> button in <i>Camera optics</i> section.  Picture 7 - Follow the instructions on the screen. Place a 15mm tube in position 1 of <i>Samples Carousel</i>. Ensure the barcode of the label is facing outwards.	--- New window appears (Picture 8). ---

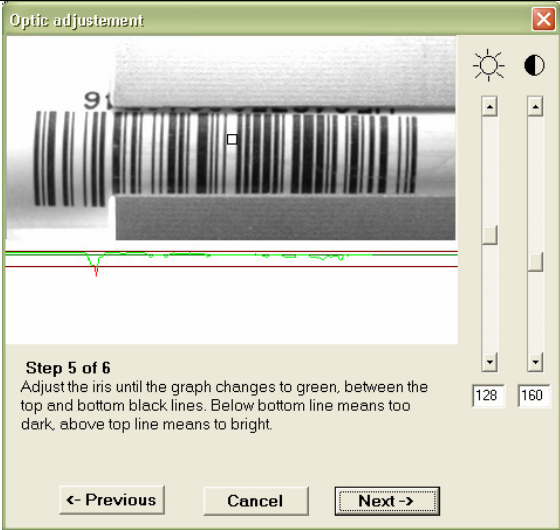
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Process	Expected Results
 Picture 8 12. Click on <i>Next</i>. 13. Centre the blue square on the barcode. The blue cross has to be in the centre of the barcode.  Picture 9 14. Click on <i>Next</i>. 15. Open the iris until the image gets too bright. Move the focus until the graph gets to its maximum position. Depending on many factors the graph may get higher or lower, but it is possible to find only one peak.	New window appears (Picture 9). --- New window appears (Picture 10). ---

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Process	Expected Results
 Picture 10 16. Click on <i>Next</i>. 17. Place the centre of the square to the brightest part of the label. Avoid the reflection coming from the LED's.  Picture 11 18. Click on <i>Next</i>. 19. Move the iris until the graph turns to green. When the graph is red color the image is too bright (on top of upper horizontal line) or too dark (below the lower horizontal line)	New window appears (Picture 11). --- New window appears (Picture 12). ---

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Process	Expected Results
 Picture 12 20. Click on <i>Next</i>. New window appears. Wait until <i>Finish</i> button is available and then click it. 21. Lock the two camera rings. Checking Samples camera adjustment 22. Place a 15mm tube in any position of the Samples Carousel. 23. Click on <i>Read Sample</i> button in <i>Reader</i> tab. 24. Type the desired position in the Carousel and click OK. 25. Click on <i>Read</i> button. The sample tube will be read. Ensure that: - The camera is able to read user's barcodes. - The instrument is able to recognise tube size. 26. Repeat steps 22 to 25 with 12 and 10 mm tubes.	--- --- --- New window appears requesting position in Carousel. New window appears. --- ---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	06/05/2005	Included in step 8 camera distance dependence by positioning.	<i>Document REGD-0000810_v3 replaces document 57038V0M001.</i>
4.0	21/09/2010	Updated all the pictures.	The document is updated due to the implementation of new functionalities.
4.0	See document header.	Document REGD-0000810_v4 replaces document REGD-0000810_v3.	Document version changed.
5.0	See document header.	Added previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
5.0	See document header.	Document REGD-0000810_5 replaces document REGD-0000810_v4.	Document version changed.
5.0	See document header.	Procedure updated.	Step 26 corrected.
5.0	See document header.	Document title.	Commercial reference substituted by project reference.
5.0	See document header.	Document header.	Increase information to easy document search.